

5.6 Quadratic and Cubic Equations

The roots of simple algebraic equations can be viewed as being functions of the equations' coefficients. We are taught these functions in elementary algebra. Yet, surprisingly many people don't know the right way to solve a quadratic equation with two real roots, or to obtain the roots of a cubic equation.

There are two ways to write the solution of the *quadratic equation*

$$ax^2 + bx + c = 0 \quad (5.6.1)$$

with real coefficients a, b, c , namely

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (5.6.2)$$

and

$$x = \frac{2c}{-b \pm \sqrt{b^2 - 4ac}} \quad (5.6.3)$$

If you use *either* (5.6.2) or (5.6.3) to get the two roots, you are asking for trouble: If either a or c (or both) is small, then one of the roots will involve the subtraction of b from a very nearly equal quantity (the discriminant); you will get that root very inaccurately. The correct way to compute the roots is

$$q \equiv -\frac{1}{2} \left[b + \operatorname{sgn}(b) \sqrt{b^2 - 4ac} \right] \quad (5.6.4)$$

Then the two roots are

$$x_1 = \frac{q}{a} \quad \text{and} \quad x_2 = \frac{c}{q} \quad (5.6.5)$$

If the coefficients a, b, c , are complex rather than real, then the above formulas still hold, except that in equation (5.6.4) the sign of the square root should be chosen so as to make

$$\operatorname{Re}(b^* \sqrt{b^2 - 4ac}) \geq 0 \quad (5.6.6)$$

where Re denotes the real part and asterisk denotes complex conjugation.

Apropos of quadratic equations, this seems a convenient place to recall that the inverse hyperbolic functions $\sinh^{-1}$ and $\cosh^{-1}$ are in fact just logarithms of solutions to such equations

$$\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1}) \quad (5.6.7)$$

$$\cosh^{-1}(x) = \pm \ln(x + \sqrt{x^2 - 1}) \quad (5.6.8)$$

Equation (5.6.7) is numerically robust for $x \geq 0$. For negative x , use the symmetry $\sinh^{-1}(-x) = -\sinh^{-1}(x)$. Equation (5.6.8) is of course valid only for $x \geq 1$.

For the *cubic equation*

$$x^3 + ax^2 + bx + c = 0 \quad (5.6.9)$$

with real or complex coefficients a, b, c , first compute

$$Q \equiv \frac{a^2 - 3b}{9} \quad \text{and} \quad R \equiv \frac{2a^3 - 9ab + 27c}{54} \quad (5.6.10)$$

If Q and R are real (always true when a, b, c are real) and $R^2 < Q^3$, then the cubic equation has three real roots. Find them by computing

$$\theta = \arccos(R/\sqrt{Q^3}) \quad (5.6.11)$$

in terms of which the three roots are

$$\begin{aligned} x_1 &= -2\sqrt{Q} \cos\left(\frac{\theta}{3}\right) - \frac{a}{3} \\ x_2 &= -2\sqrt{Q} \cos\left(\frac{\theta + 2\pi}{3}\right) - \frac{a}{3} \\ x_3 &= -2\sqrt{Q} \cos\left(\frac{\theta - 2\pi}{3}\right) - \frac{a}{3} \end{aligned} \quad (5.6.12)$$

(This equation first appears in Chapter VI of François Viète's treatise "De emendatione," published in 1615!)

Otherwise, compute

$$A = -\left[R + \sqrt{R^2 - Q^3}\right]^{1/3} \quad (5.6.13)$$

where the sign of the square root is chosen to make

$$\operatorname{Re}(R^* \sqrt{R^2 - Q^3}) \geq 0 \quad (5.6.14)$$

(asterisk again denoting complex conjugation). If Q and R are both real, equations (5.6.13) – (5.6.14) are equivalent to

$$A = -\operatorname{sgn}(R) \left[|R| + \sqrt{R^2 - Q^3}\right]^{1/3} \quad (5.6.15)$$

where the positive square root is assumed. Next compute

$$B = \begin{cases} Q/A & (A \neq 0) \\ 0 & (A = 0) \end{cases} \quad (5.6.16)$$

in terms of which the three roots are

$$x_1 = (A + B) - \frac{a}{3} \quad (5.6.17)$$

(the single real root when a, b, c are real) and

$$\begin{aligned} x_2 &= -\frac{1}{2}(A + B) - \frac{a}{3} + i\frac{\sqrt{3}}{2}(A - B) \\ x_3 &= -\frac{1}{2}(A + B) - \frac{a}{3} - i\frac{\sqrt{3}}{2}(A - B) \end{aligned} \quad (5.6.18)$$